linearAlgebra: Benchmarks

Ivan Jacob Agaloos Pesigan

1 All Tests Under tests-benchmark

```
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
#>   <bch:expr>   <bch:> <bch:>     <dbl> <bch:byt>    <dbl> <int> <dbl>   <bch:tm>
#> 1 linearAlgeb~ 16.2us 17.5us    48334.    4.06KB        0    10     0      207us
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#> 1 linearAlgebr~ 196us  199us     4908.     375KB        0    10     0     2.04ms
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#> 1 linearAlgebr~ 762us  773us     1269.    1.65MB        0    10     0     7.88ms
#> # i 4 more variables: result <list>, memory <list>, time <list>, gc <list>
#> # A tibble: 2 x 13
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#> 1 linearAlgeb~ 12.1us 13.2us    73061.    8.05KB        0    20     0      274us
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#> 1 linearAlgeb~ 5.42us 6.38us   141909.    8.11KB        0    10     0     70.5us
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#>   <bch:expr>   <bch:> <bch:>     <dbl> <bch:byt>    <dbl> <int> <dbl>   <bch:tm>
#> 1 linearAlgeb~ 5.97us 6.57us   134724.    8.77KB        0    10     0     74.2us
#> # i 4 more variables: result <list>, memory <list>, time <list>, gc <list>
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
```

```
#> [[1]][[1]]$value[[1]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
#>   <bch:expr>   <bch:> <bch:>     <dbl> <bch:byt>    <dbl> <int> <dbl>   <bch:tm>
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#>
#> [[1]][[1]]$value[[2]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
#>   <bch:expr>    <bch> <bch:>     <dbl> <bch:byt>    <dbl> <int> <dbl>   <bch:tm>
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#>
#> [[1]][[1]]$value[[3]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
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#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
```

```
#> [[1]][[2]]$value
#> [[1]][[2]]$value[[1]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
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#> [[1]][[2]]$value[[2]]
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#> [1] TRUE
#>
#>
```

```
#> [[1]][[3]]
#> [[1]][[3]]$value
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#> # A tibble: 2 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
#>   <bch:expr>   <bch:> <bch:>     <dbl> <bch:byt>    <dbl> <int> <dbl>   <bch:tm>
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#>
```

```
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
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```
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> # A tibble: 1 x 13
#>   expression      min median `itr/sec` mem_alloc `gc/sec` n_itr  n_gc total_time
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#> [1] TRUE
```

Environment

```
ls()


#> [1] "root"
```

Class

```
#> [[1]]
#> [1] "root_criterion"
```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/